Module 4: Sampling and Quantization

Why Digitize Analog sources:

1) Digital systems are less sensitive to Noise than analog systems.
2) In Digital system it is easier to integrate different services like audio and video in same transmission scheme.
3) Transmission scheme can be relatively independent of source.
   Eg: voice at 10KbPS and computer data at 10Kbps can be sent at same scheme.
4) Digital signal circuitry are easy to handle and repeat.
5) multiplexing of analog signals is difficult and not efficient.
6) Analog data transmission do not have efficient error detection and correction technique.

Sampling process:

Statement of sampling Theorem.

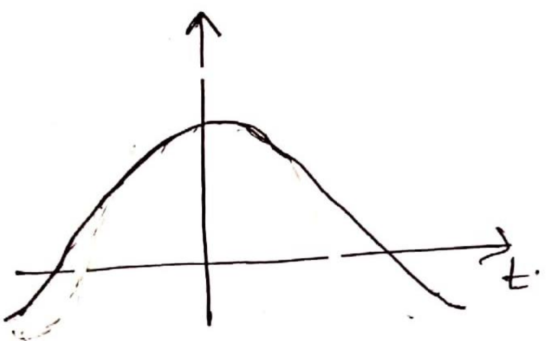
1) A band limited signal of finite energy which has all frequencies lesser than W Herz, it completely described by specifying the values of the signal at instants of time separated by $\frac{1}{2W}$ seconds.
2) A band limited signal of finite energy which has frequency components ~~to~~ lesser than W Hertz, may be completely recovered from the knowledge of its samples taken at the rate of 2W samples per sec.

Sampling theorem for low pass signals:

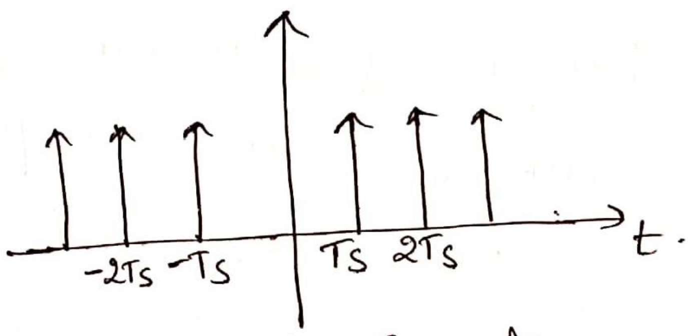
"A bandlimited signal have frequency 'w' hz can be completely described by its samples if the sampling frequency 'f_s' is greater than or equal to twice the highest frequency of message signal.

$\therefore f_s \geq 2W$.

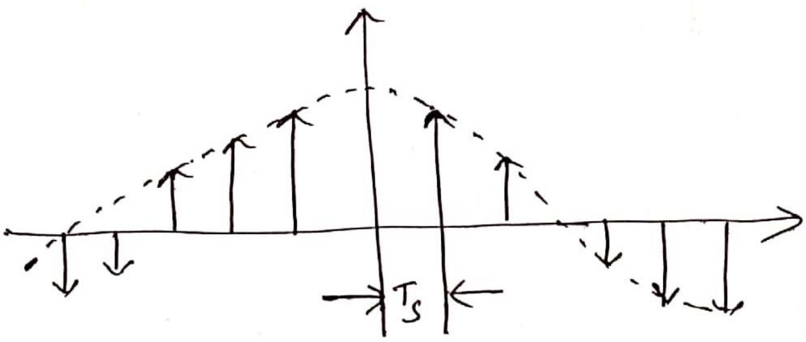
Time Domain analysis:



Analog signal.



periodic impulse



Discrete time signal.

Proof: Let $g(t)$ be continuous wave signal which is sampled by pulses.

Sampled o/p $= g_\delta(t) = g(t)\, s_\delta(t)$. —— ①

periodic impulse train $s_\delta(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$. —— ②

substituting ② in ①

$$g_\delta(t) = g(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s).$$

$$g_\delta(t) = \sum_{n=-\infty}^{\infty} g(nT_s)\, \delta(t - nT_s).$$

where $g(nT_s)$ is basically $g(t)$ sampled to $t = nT_s$, $n = 0, \pm 1, \pm 2 \ldots\ldots$

Frequency domain:

Taking fourier transform of eq ①.

$G_\delta(f) = G(f) * s_\delta(f)$. —— ④

where $s_\delta(f) = f_s \sum_{n=-\infty}^{\infty} \delta(f - nf_s)$ —— ⑤.

substituting ⑤ in ④

$G_\delta(f) = G(f) * f_s \sum_{n=-\infty}^{\infty} \delta(f - nf_s) \Rightarrow f_s \sum_{n=-\infty}^{\infty} G(f) * \delta(f - nf_s)$

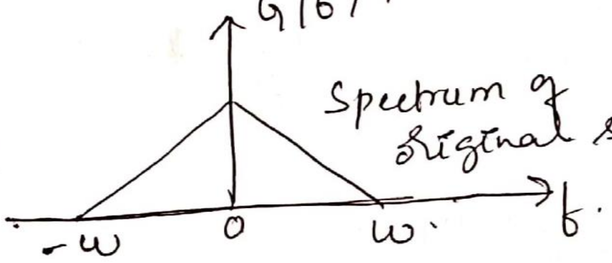
wkt · convolution property:

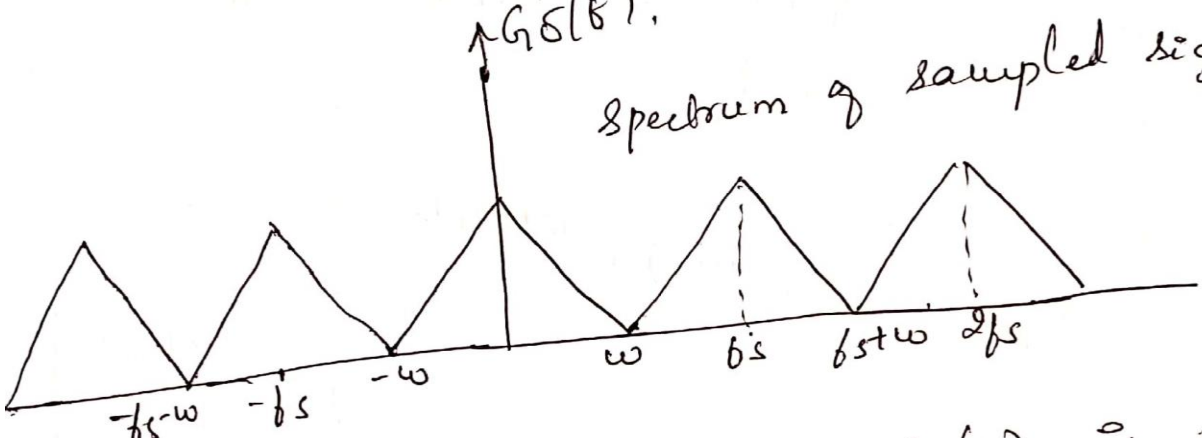
$$G(f) * \delta(f - nf_s) = G(f - nf_s).$$

$$\boxed{\therefore G_\delta(f) = f_s \sum_{n=-\infty}^{\infty} G(f - nf_s).} \quad \text{—— ⑥}$$

By shifting property

$G_\delta(f) = \ldots f_s G(f_s + 2f_s) + f_s G(f + f_s) + f_s G(f) + f_s G(f - f_s) + f_s G(f - 2f_s).$





The above shows that G(f) is placed at $\pm f_s$, $\pm 2f_s$, $\pm 3f_s$ - - - .

It means G(f) is periodic in f_s.

Let $f_s = 2W$.

$G_\delta(f) = f_s\, G(f)$ for $-W \leq f \leq W$ & $f_s = 2W$.

$G(f) = \dfrac{G_\delta(f)}{f_s}$

DTFT: $G(\Omega) = \sum\limits_{n=-\infty}^{\infty} g(n)\, e^{-j\Omega n}$.

$G(f) = \sum\limits_{n=-\infty}^{\infty} g(n)\, e^{-j2\pi f n}$

where f → frequency of discrete time signal

Replacing $g(n)$ by $g(nT_s)$.

$$G_\delta(f) = \sum_{n=-\infty}^{\infty} g(nT_s)\, e^{-j2\pi \frac{f}{f_s} n}$$

$$T_s = \frac{1}{f_s}.$$

$$G_\delta(f) = \sum_{n=-\infty}^{\infty} g(nT_s)\, e^{-j2\pi f n T_s}.$$

~~when the spectrum of ... above equation is passed through LPF~~

we have.

$$G(f) = \frac{1}{f_s} G_\delta(f).$$

$$= \frac{1}{f_s} \sum_{n=-\infty}^{\infty} g(nT_s)\, e^{-j2\pi f n T_s}.$$

$$\therefore \text{IFT of } g(t) = \text{IFT}\left[\frac{1}{f_s} \sum_{n=-\infty}^{\infty} g(nT_s)\, e^{-j2\pi f n T_s}\right]$$

Reconstruction of $g(t)$:

IFT of the above equation is

$$g(t) = \int_{-\infty}^{\infty} \left[\frac{1}{f_s} \sum_{n=-\infty}^{\infty} g(nT_s)\, e^{-j2\pi f n T_s}\right] e^{j2\pi f t}\, df.$$

Since $-w \le f \le w$,

$$= \int_{-w}^{w} \frac{1}{f_s} \sum_{n=-\infty}^{\infty} g(nT_s)\, e^{-j2\pi f n T_s} \cdot e^{j2\pi f t}\, df$$

Interchanging order of summation & integration.

$$g(t) = \sum_{n=-\infty}^{\infty} g(nT_s) \frac{1}{f_s} \int_{-w}^{w} e^{j2\pi f(t-nT_s)} \, df.$$

$$= \sum_{n=-\infty}^{\infty} g(nT_s) \frac{1}{f_s} \left[\frac{e^{j2\pi f(t-nT_s)}}{j2\pi(t-nT_s)} \right]_{-w}^{w}$$

$$= \sum_{n=-\infty}^{\infty} g(nT_s) \frac{1}{f_s} \left\{ \frac{e^{j2\pi w(t-nT_s)} - e^{-j2\pi w(t-nT_s)}}{j2\pi(t-nT_s)} \right\}.$$

$$= \sum_{n=-\infty}^{\infty} g(nT_s) \frac{1}{f_s} \left\{ \frac{\sin 2\pi w(t-nT_s)}{\pi(t-nT_s)} \right\} \quad \left[\because \sin\theta = \frac{1}{2j}\left\{ \exp(j\theta) - \exp(-j\theta) \right\} \right]$$

$$= \sum_{n=-\infty}^{\infty} g(nT_s) \left\{ \frac{\sin 2\pi w(t-nT_s)}{\pi(f_s t - f_s nT_s)} \right\}. \qquad f_s = 2w$$

$$T_s = \frac{1}{f_s} = \frac{1}{2w}.$$

$$= \sum_{n=-\infty}^{\infty} g(nT_s) \frac{\sin\pi\left[(2wt - 2wnT_s)\right]}{\pi\left(2wt - f_s n \frac{1}{f_s}\right)}$$

$$= \sum_{n=-\infty}^{\infty} g(nT_s) \frac{\sin\pi\left[2wt - 2wn\frac{1}{2w}\right]}{\pi(2wt - n)}$$

$$g(t) = \sum_{n=-\infty}^{\infty} g(nT_s) \frac{\sin \pi (2wt - n)}{\pi (2wt - n)}.$$

$$g(t) = \sum_{n=-\infty}^{\infty} g(nT_s) \; \text{sinc}(2wt - n) \qquad \left[\frac{\sin \pi \theta}{\pi \theta} = \text{sinc}(\theta) \right.$$

By Expanding the above equation :

$$\therefore g(t) = \ldots\ldots + g(-2T_s) \,\text{sinc}(2wt + 2) + g(-T_s) \,\text{sinc}(2wt + 1) + g(0) \,\text{sinc}(2wt) + g(T_s) \,\text{sinc}(2wt - 1) + \ldots$$

Comments :

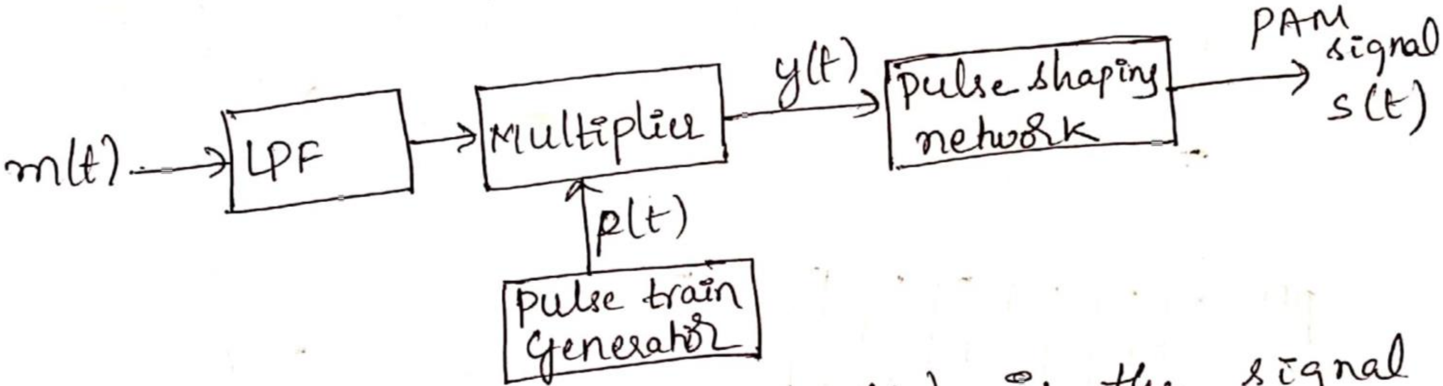
1) The sample $g(nT_s)$ are weighed by sinc function.
2) Sinc function is the interpolating function
3) Reconstruction of $g(t)$ by LPF

i.e. when interpolated signal of above equation is passed through the low pass filter of b/w $-w \le f \le w$, the reconstruction of wave is obtained.

Pulse-Amplitude Modulation: [PAM]

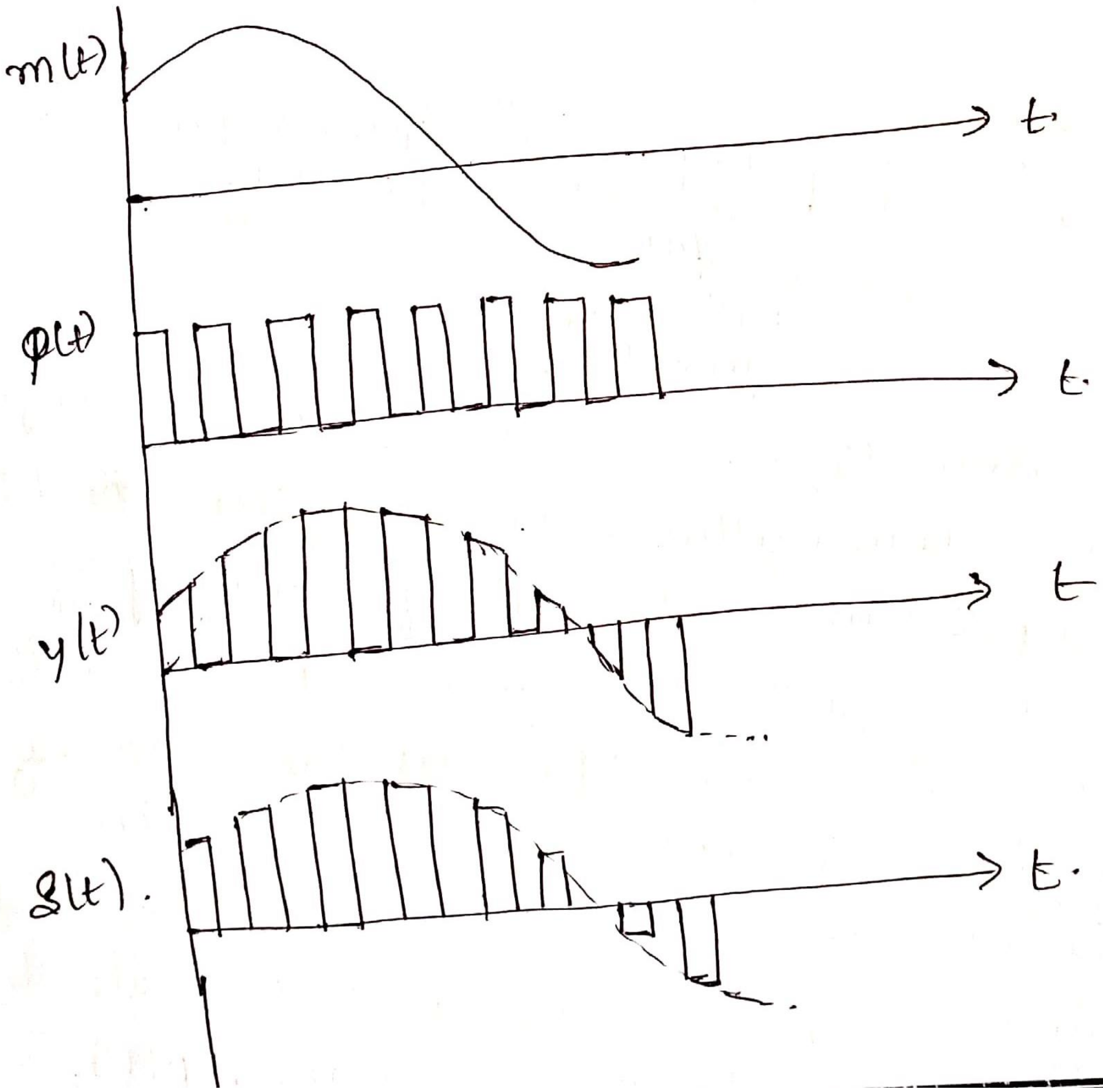
- The amplitude of pulse is directly proportional to amplitude of the modulating signal at the sampling instant keeping width and position of pulse constant.

Generation of PAM:



- The modulating signal m(t) is the signal to be transmitted. It is given to LPF to band limit. The cutoff frequency of LPF is equal to highest frequency (fm) present in m(t). LPF avoids aliasing.
- The band limited signals is then sampled at the multiplier. The multiplier samples m(t) with the help of pulse train generator which produces pulse train p(t).

- The multiplication of m(t) and p(t) produces the PAM signal y(t), the top of the pulses are varied according to amplitude of m(t).
- The pulse shaping n/w produces flat top pulses which is a required PAM signal as shown below.



* Let $s(t)$ denote the sequence of flat-top samples of PAM signal, and it is expressed as.

$$s(t) = \sum_{n=-\infty}^{\infty} g(nT_s)\, p(t-nT_s).$$

where

$g(nT_s)$ – is the sample value of $g(t)$ obtained at time $t = nT_s$

~~[illegible]~~.

T_s – sampling period.

$p(t)$ – standard rectangular pulse train of duration T.

Advantage of PAM:

1. It is a base for all the digital modulation technique.

Disadvantage of PAM:

1) Due to Nyquist Criteria, it requires high bandwidth for transmission.
2) Since, amplitude keep varying, so there is noise associated with it.

Detection of PAM signal:

The original message signal m(t) is obtained by passing PAM signal to the reconstruction filter followed by equilizer.

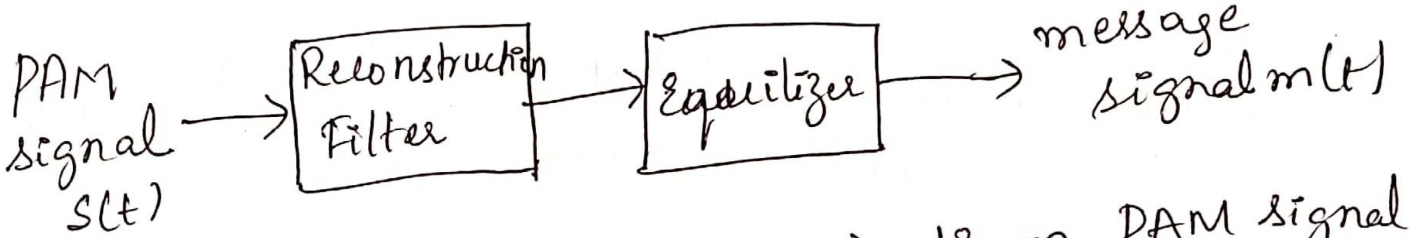


fig (b): Recovering m(t) from PAM signal.

Time Division Multiplexing [TDM]

- TDM is a method of transmitting and receiving independent signals over a common channel by means of synchronised switches at each end of transmission line so that each signal appears on the line only a fraction of time in an alternating pattern.

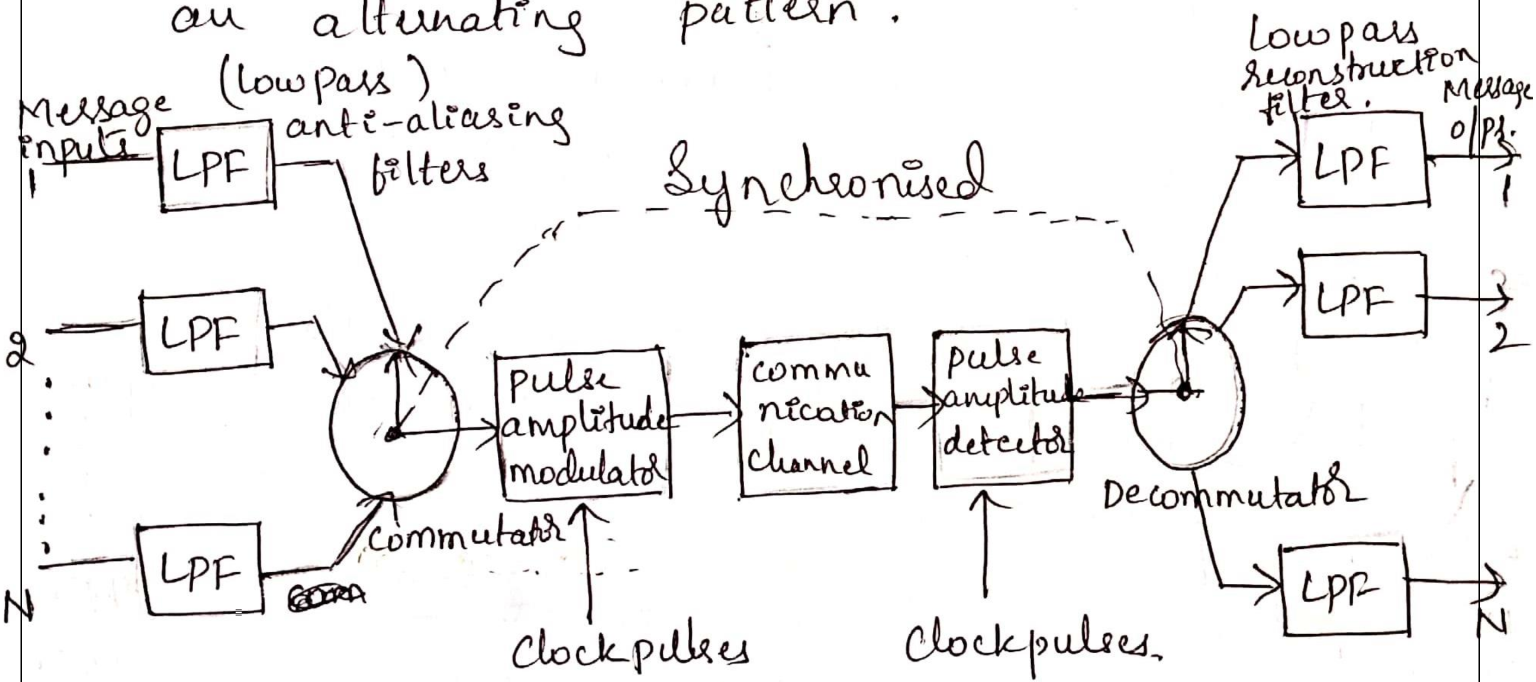


fig: Block diagram of TDM system.

- In the block diagram, each input message signal is first bandlimited by LPF to remove higher unwanted frequencies.
- The output of the ~~ϕ~~ anti aliasing filters are then fed to a commutator, which is an electronic switching circuitry.
- The function of commutator is as follows:
  1) To take a narrow sample of each of the 'N' samples of input at rate of $f_s \geq 2W$.
  2) To sequentially interleave (multiplex) these 'N' samples inside a sampling interval $T_s = \frac{1}{f_s}$.
- The multiplexed signal is then applied to a pulse amplitude modulator whose purpose is to transform the multiplexed signal into a form suitable for transmission over a common channel.

- At the receiving end, the pulse amplitude demodulator performs the reverse operation of PAM and the decommutator distributes the signals to the appropriate low pass reconstruction filters. The decommutator operates in synchronisation with the commutator.

Pulse - position Modulation: (PPM).

- In pulse-duration modulation (PDM), the samples of the message signal are used to vary the duration of the individual pulses. This form of modulation is also referred to as pulse-width modulation or pulse-length modulation.

- In PPM, the position of a pulse relative to its unmodulated time of occurance is varied in accordance with the message signal as shown in fig (d) for the case of sinusoidal modulation.

Let T_s denote the sample duration. Using the sample $m(nT_s)$ of a message signal $m(t)$ to modulate the position of the n^{th} pulse, we obtain the PPM signal

$$s(t) = \sum_{n=-\infty}^{\infty} g(t - nT_s - k_p m(nT_s)).$$

where k_p is the sensitivity of the pulse position modulator.

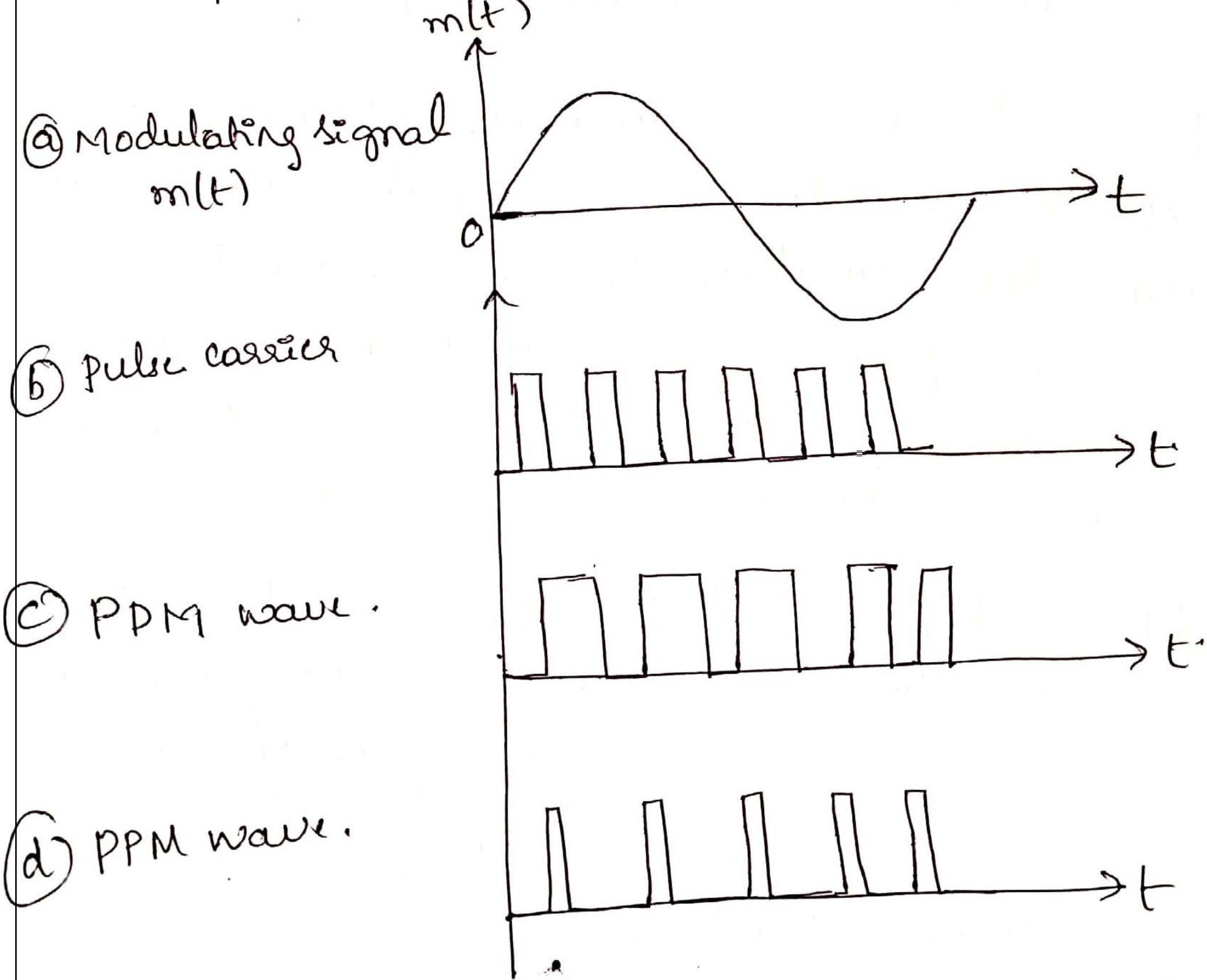
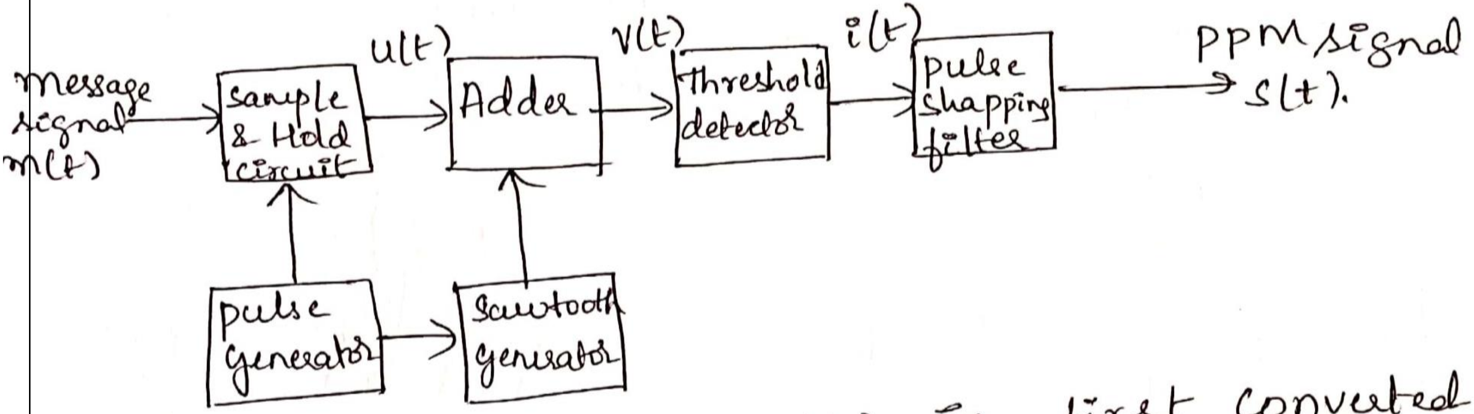


Fig: Illustrating two different forms of pulse-time modulation

Generation of PPM waves:

- The PPM signal which is generated is shown in fig.



- The message signal m(t) is first converted into a PAM signal by means of a sample & Hold circuit, generating a staircase wave u(t) which is shown in fig (b). for the message signal m(t) shown in fig (a).
- Next, the signal u(t) is added to a sawtooth wave, yielding the combined V(t).
- The combined V(t) is applied to a threshold detector that produces a very narrow pulse each time V(t) crosses zero in the

-ve going direction.

- The resulting sequence of "impulses" $i(t)$ is shown in fig (e).
- Finally, the PPM signal $s(t)$ is generated by using this sequence of impulses to excite a filter whose impulse response is defined by the standard pulse $g(t)$.

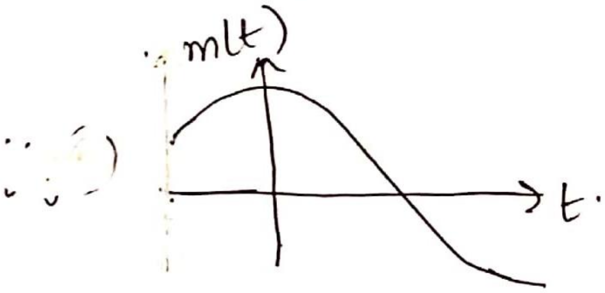


fig (a) message signal.

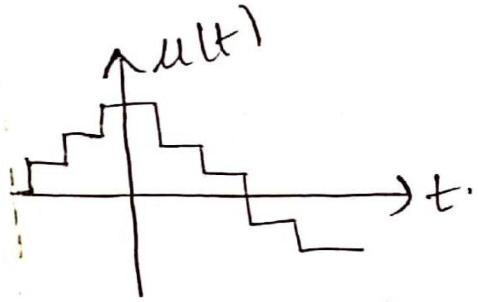


fig (b) Staircase signal

fig (c) Sawtooth wave.

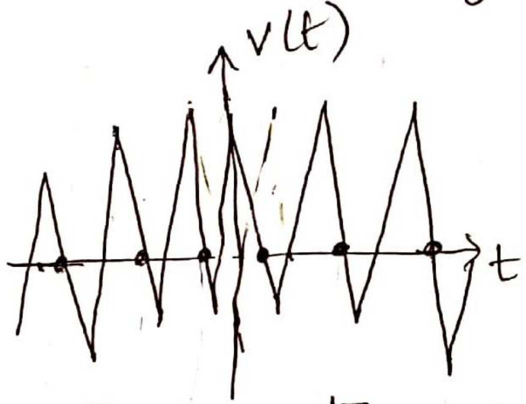


fig (d) composite wave obtained by adding (b) & (c).

fig (e) Sequence of "Impulses" used to generate the PPM signal.

Detection of PPM waves:

Consider a PPM wave $s(t)$ with uniform sampling and assume that the message signal $m(t)$ is strictly bandlimited.

The operation of one type of PPM receiver may proceed as follows:

1) Convert the received PPM wave into a PDM wave with the same modulation.
2) Integrate this PDM using a device with a finite integration time, thereby computing the area under each pulse of the PDM wave.
3) Sample the o/p of the integrator at a uniform rate to produce a PAM wave, whose pulse amplitudes are proportional to the signal samples $m(nT_s)$ of the original PPM wave $s(t)$.
4) Finally, demodulate the PAM wave to recover the message signal $m(t)$.

Noise in pulse-position Modulation:

- In a PPM system, the transmitted information is contained in the relative positions of the modulated pulses. The presence of additive noise affects the performance of such a system.
- The o/p signal to noise ratio, assuming a full-load sinusoidal modulation is therefore

$$(SNR)_o = \frac{\pi^2 B_T T_s^2 A^2}{32 N_0} \quad \text{——— (1)}$$

- The average noise power in a message bandwidth 'W' is equal to WN_0. The channel signal to noise ratio is therefore

$$(SNR)_c = \frac{3A^2}{4 T_s B_T W N_0} \quad \text{——— (2)}$$

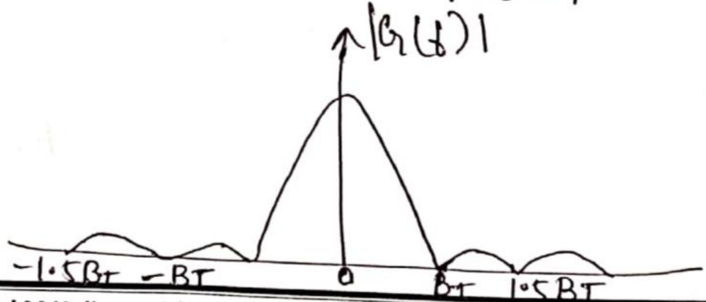


fig: Amplitude spectrum of a raised cosine pulse.

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where T_s = sampling period.
B_T = Bandwidth.
N_0 = power spectral.

- Thus, the figure of merit of a PPM system using a raise ~~cos~~ cosine pulse is a follows:

$$\text{Figure of Merit} = \frac{(SNR)_o}{(SNR)_c} = \frac{\pi^2 B_T T_s^2 A^2}{32 N_0} \times \frac{4 T_s B_T W N_0}{3A^2}$$

$$\therefore \boxed{\text{Figure of Merit} = \frac{\pi^2 B_T^2 T_s^3 W}{24}}$$

Quantization process:

- The process of transforming sampled amplitude $m(nT_s)$ of a message signal $m(t)$ into a discrete amplitude $v(nT_s)$ (levels) is reffered to as <u>quantization</u>.

Continuous Sample (m) $\longrightarrow$ $\boxed{\text{Quantizer } g(\cdot)}$ $\longrightarrow$ Discrete sample (v).

- Quantization can be uniform or Nonuniform.
- The quantizer characteristics can be mid-thread or mid-rise type.

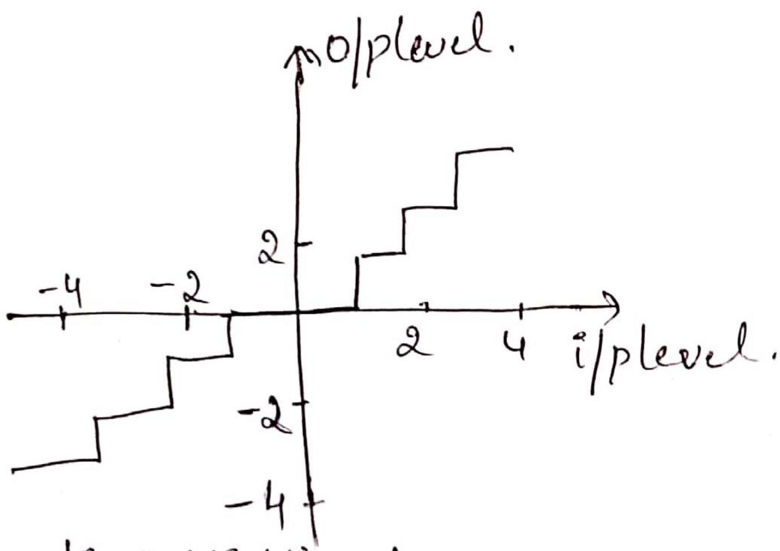


fig: Midtread.

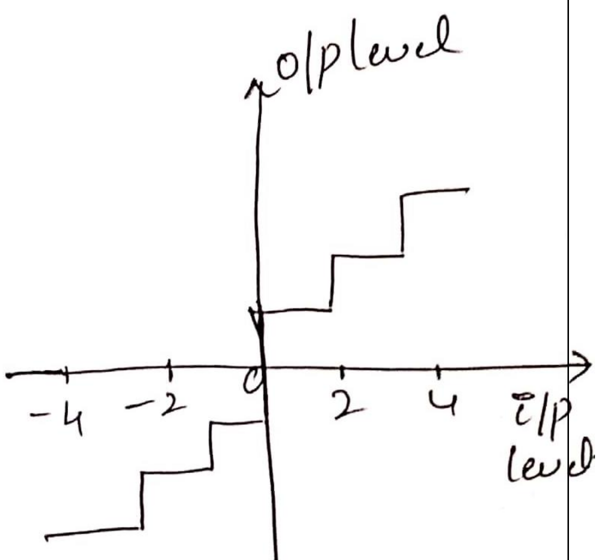


fig: Mid-rise.

Quantization Noise:

- The use of quantization introduces an error defined as the difference b/w the i/p signal m and o/p signal v. This error is called quantization noise.
- Let the quantization error be denoted by the random variable Q of sample value q. we may thus write

$$q = m - v \quad \text{--- (1)}$$

or

$$Q = M - V.$$

- fig. illustrates a typical variation of the quantization noise as a function of time assuming the use of a uniform quantizer of the mid tread type.

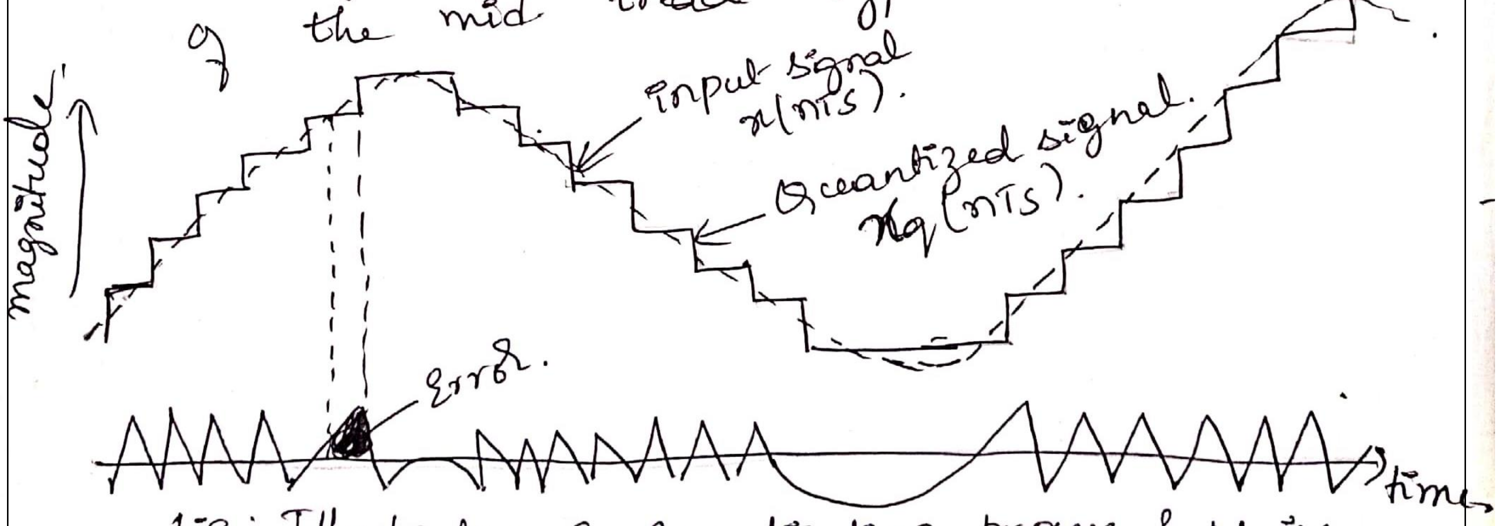


fig: Illustration of quantization process & Noise.

* consider then an i/p 'm' of continuous amplitude in the range $(-m_{max}, m_{max})$ then, the step-size of the quantizer is given by

$$\Delta = \frac{2m_{max}}{L} \quad \text{--- (2)}$$

where L is the total number of representation levels.

* Now the probability density function of the quantization error Q as follows.

$$f_Q(q) = \begin{cases} \frac{1}{\Delta} & -\frac{\Delta}{2} < q \leq \frac{\Delta}{2} \\ 0 & \text{otherwise} \end{cases}$$

* Now the variance of Quantization error is σ_Q^2

$$\sigma_Q^2 = \int_{-\Delta/2}^{\Delta/2} q^2 f_Q(q)\, dq$$

$$= \int_{-\frac{\Delta}{2}}^{\frac{\Delta}{2}} q^2 \frac{1}{\Delta}\, dq \qquad \left(\because f_Q(q) = \frac{1}{\Delta}\right)$$

$$\sigma_Q^2 = \frac{1}{\Delta}\left[\frac{q^3}{3}\right]_{-\Delta/2}^{\Delta/2}$$

$$= \frac{1}{3\Delta}\left[\left(\frac{\Delta}{2}\right)^3 - \left(-\frac{\Delta}{2}\right)^3\right].$$

$$= \frac{1}{3\Delta}\left[\frac{\Delta^3}{8} - \frac{(-\Delta^3)}{8}\right]$$

$$= \frac{1}{3\Delta}\left[\frac{\Delta^3}{8} + \frac{\Delta^3}{8}\right]$$

$$= \frac{1}{3\cancel{\Delta}}\left[\frac{\cancel{2}\Delta^{\cancel{3}}}{\cancel{8}4}\right]$$

$$\boxed{\sigma_Q^2 = \frac{\Delta^2}{12}.} \quad \text{—— (3)}$$

This is known as "Mean Squared Quantization error" or Normalized Noise power or Quantization error interms of power.

* Let us consider 'R' which denote the number of bits per sample then the quantized level is given by

$$L = 2^R \quad \text{— (4)}$$

Sub (4) in (2)

$$\Delta = \frac{2m_{max}}{2^R} \quad \text{— (5)}$$

Now sub (5) in (3)

$$\sigma_Q^2 = \frac{\left(\frac{2m_{max}}{2^R}\right)^2}{12}$$

$$= \frac{4 m^2_{max}}{2^{2R}} \times \frac{1}{12}$$

$$\boxed{\sigma_Q^2 = \frac{1}{3} m^2_{max} 2^{-2R}} \quad \text{— (6)}$$

Let 'P' denote the average power of message signal m(t), we may express the output signal to noise ratio of a uniform quantizer as.

$$(SNR)_0 = \frac{P}{\sigma_Q^2}$$

$$= \frac{P}{\frac{1}{3} m_{max}^2 2^{-2R}}.$$

$$\boxed{(SNR)_0 = \left(\frac{3P}{m^2_{max}}\right) 2^{2R}} \quad \text{---} \; (7)$$

Equation (7) shows that the o/p signal-to-noise ratio of the quantizer increases exponentially with increasing number of bits per sample, R.

Pulse-code Modulation:

* In pulse code modulation (PCM), a message signal is represented by a sequence of coded pulses, which is accomplished by representing the signal in discrete form in both time and amplitude.
* The basic operations performed in the transmitter of a PCM system are sampling, quantizing and encoding as shown in fig ⓐ. The lowpass filter prior to sampling is included to prevent aliasing of the message signal.
The quantizing & encoding operations are usually performed in same circuit, which is called an analog-to-digital converter.
* The basic operations in the receiver are generation of impaired signals, decoding & reconstruction of the train of quantized samples as shown in fig ⓒ. Regeneration also occurs at intermediate points along

the transmission path as necessary as indicated in fig ⓑ.

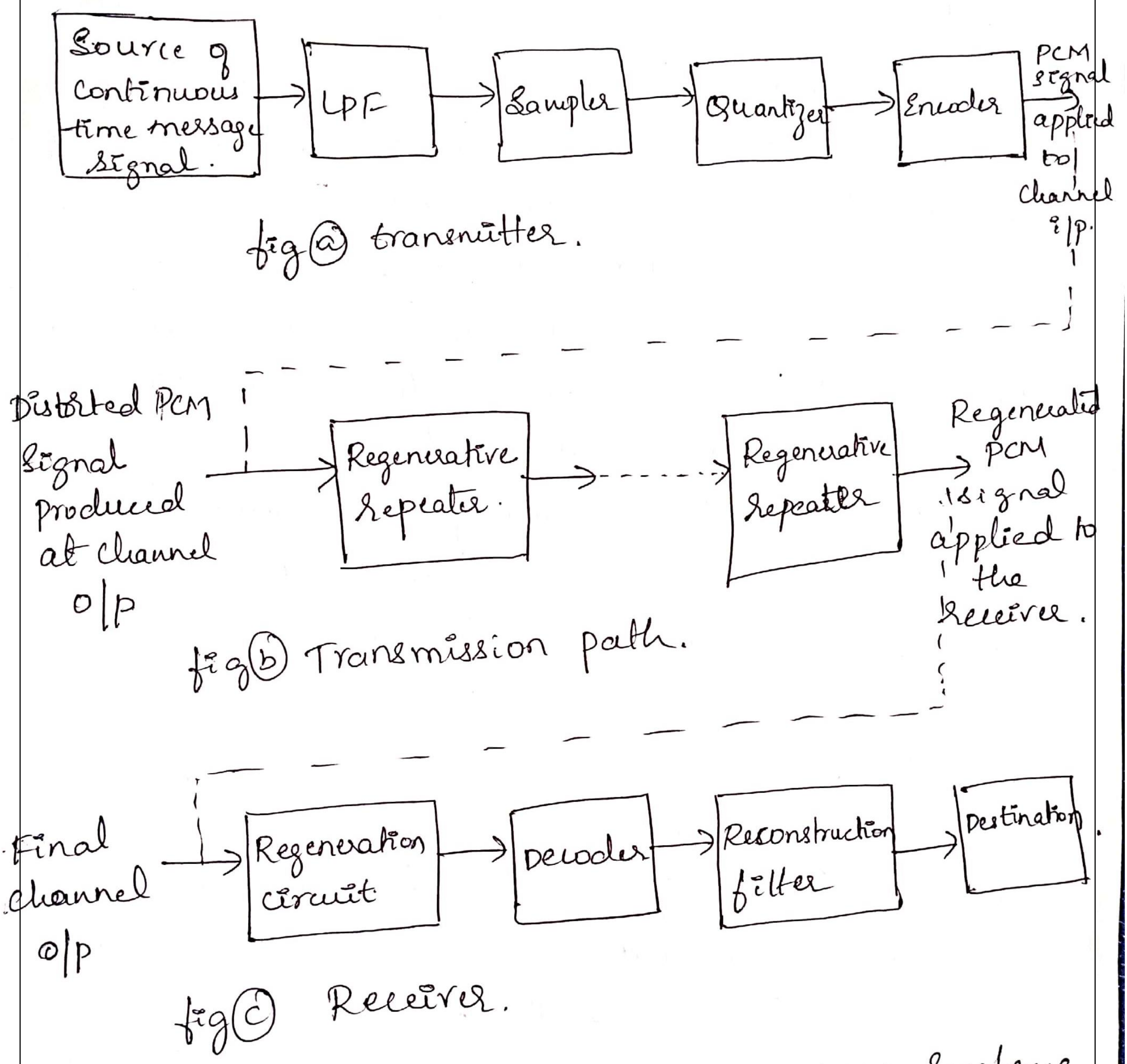


fig: The basic elements of a PCM Systems.

Sampling:

The incoming message signal is sampled with a train of narrow rectangular pulses so as to closely approximate the instantaneous sampling process. In order to ensure perfect reconstruction of the message signal at the receiver, the sampling rate must be greater than or equal to the highest frequency component 'W' of the message signal in accordance with the sampling theorem.

$$f_s \geq 2W.$$

Quantization:

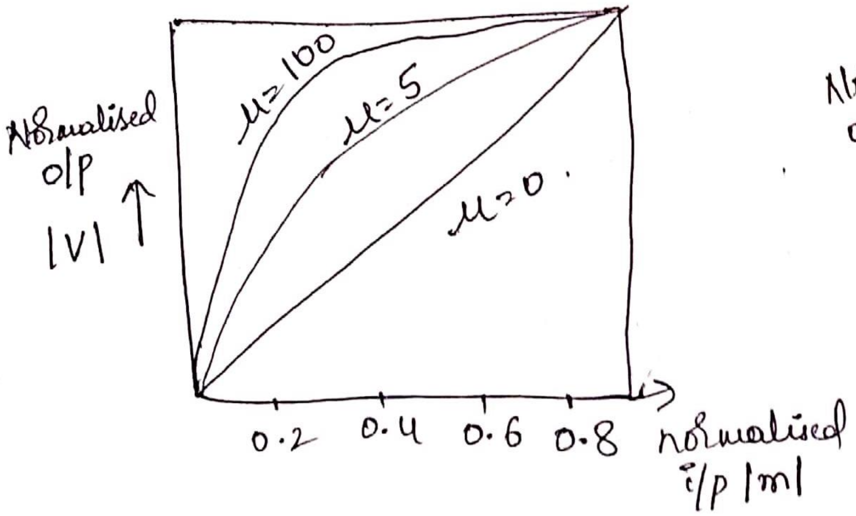
- The sampled version of the message signal is then quantized, thereby providing a new representation of the signal that is discrete in both time & amplitude
- For uniform quantization, we have mid-thread and mid-rise quantization and for non-uniform quantization, we have
  - compression laws (μ-law)

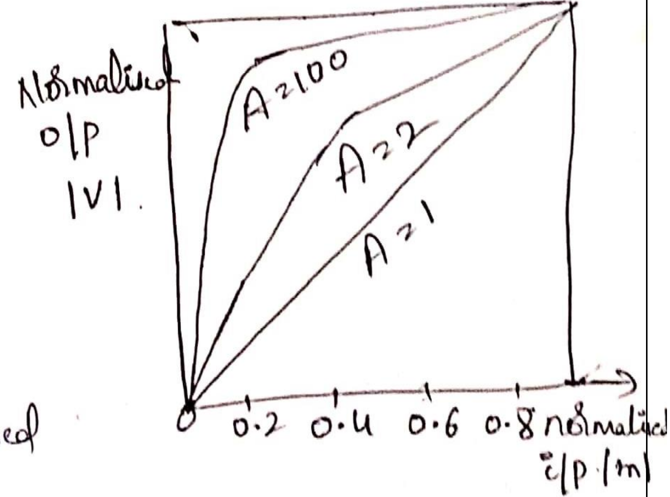
  - A law.

- The use of a non-uniform quantizer is equivalent to passing the baseband signal through ~~a~~ compressor and then applying the compressed signal to a uniform quantizer.
- A particular form of compression law that is used in practise is the so called <u>μ-law</u>, defined by

$$|v| = \frac{\log(1+\mu|m|)}{\log(1+\mu)} \quad \text{—— (1)}$$

where m and v are normalised input and output voltage, and μ is positive constant.





- Another compression law that is used in practice is the so-called A-law as show.

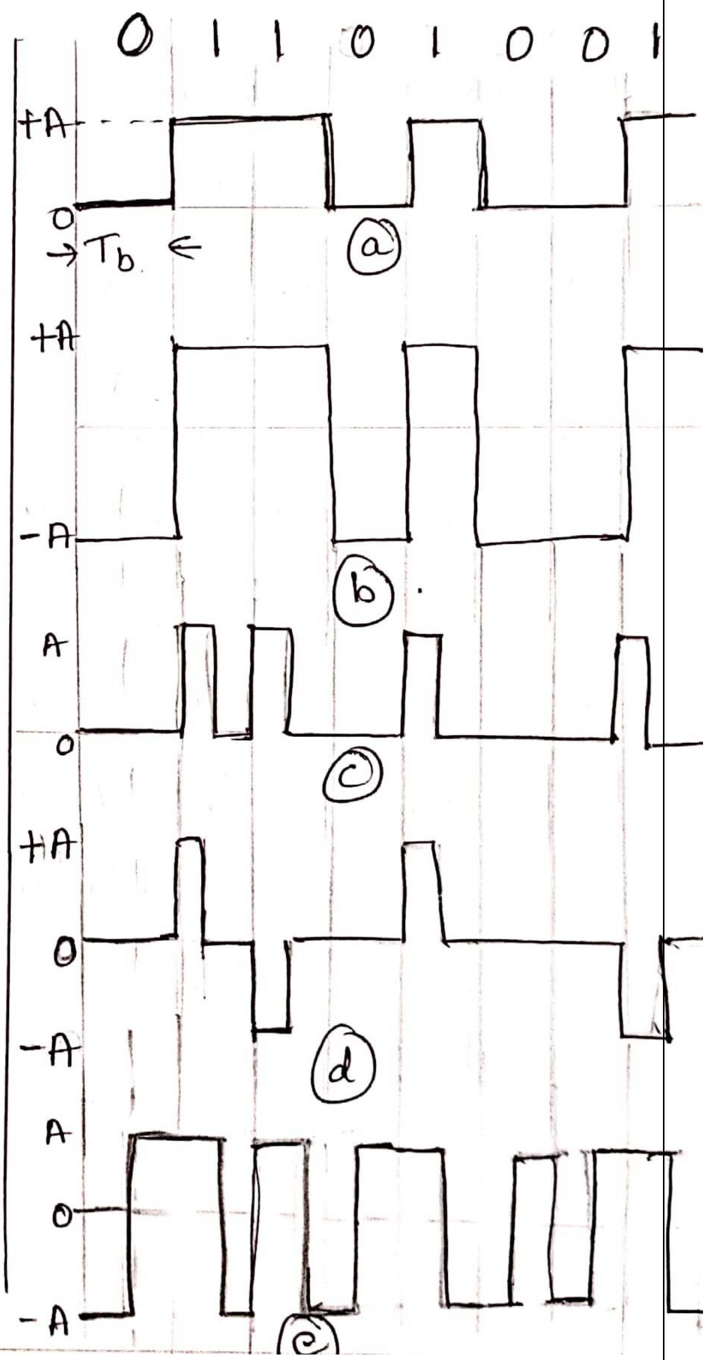
$$|v| = \begin{cases} \dfrac{A|m|}{1+\log A} & 0 \leq |m| \leq \dfrac{1}{A} \\ \dfrac{1+\log(A|m|)}{1+\log A} & \dfrac{1}{A} \leq |m| \leq 1 \end{cases}$$

Encoding:

- In combining the processes of sampling and quantizing the specification of a continuous message (baseband) signal. becomes limited to a discrete set of values but not in the form best suited to transmission over a line & radio path.
- In a binary code each symbol may be either of two distinct values or kinds, such as the presence or absence of a pulse. The two symbols of a binary code are denoted as 0 & 1.

Line Codes:

It is a line code that a binary stream of data takes on an electrical representation. The five line codes are illustrated in fig.



(1) Unipolar Non return to Zero (NRZ) Signaling:

In this line code symbol '1' is represented by transmitting a pulse of amplitude 'A' and symbol '0' is represented by switching off the pulse.

(2) polar Non-return to zero (NRZ) signaling:

In this line code, symbol 1 & 0 are represented by transmitting pulses of amplitudes +A & −A respectively.

3) unipolar Return to Zero (RZ) signalling:

Here, symbol 1 is represented by a rectangular pulse of amplitude A and half-symbol width & 0 is represented by transmitting no pulse.

4) Bipolar Return to zero (BRZ) signalling:

- This line code uses three amplitude levels as shown in fig (d). Specifically +ve & -ve pulses of equal amplitude are used alternatively for symbol 1.
- '0' for no pulses. This line code is also called alternate mark inversion (AMI) signaling

5) Split-phase (Man-chester Code)

Symbol '1' is represented a positive pulse of Amplitude 'A' followed by a negative pulse of amplitude -A, with both pulses being a half-symbol wide. For symbol '0', the polarities of these two pulses are received.

Differential Encoding:

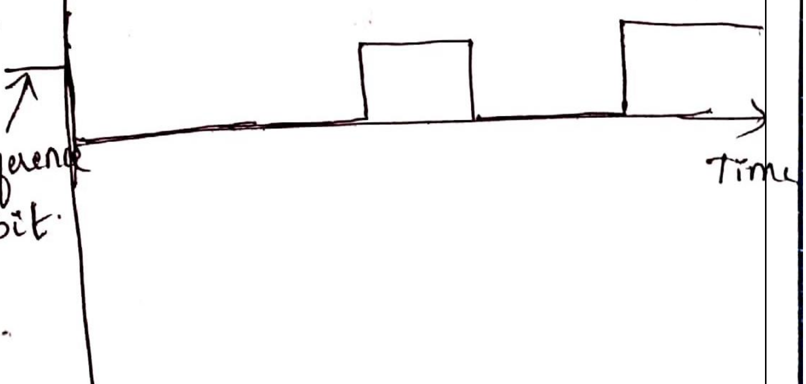
- This method is used to encode information interms of signal transitions. In particular, a transition is used to designate symbol '0' in the incoming binary data stream, while no transition is used to designate symbol '1' as shown in fig.

a) original binary data: 0 1 1 0 1 0 0 1

b) Differential encoded data '1' 0 0 0 1 1 0 1 1.

(EX-NOR).

0	0	1
0	1	0
1	0	0
1	1	1



c) wave form.

fig: Differential Encoding.

Regeneration:

The distorted PCM wave obtained from the transmitter is sent to the amplifier equilizer. The o/p of equilizer device is passed to the decision making device to decide the signal interms of 1 or 0 (coded o/p).

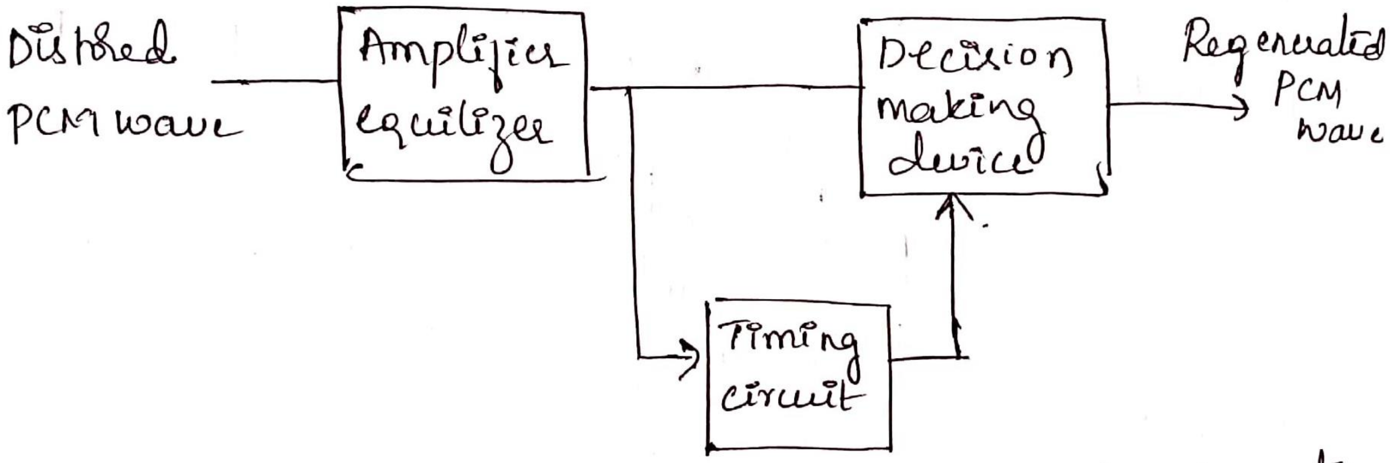


Fig: Block diagram of a regenerative repeater

Decoding:

The decoding process involves generating a pulse the amplitude of which is the linear sum of all the pulses in the codeword, with each pulse being weighted by its place value ($2^0, 2^1, 2^2 \ldots 2^{R-1}$) in the code, where 'R' is the number of bits per sample.

Filtering:

The final operation in the receiver is to recover the message signal wave by passing the decoder o/p through a lowpass reconstruction filter whose cutoff frequency is equal to the message bandwidth 'W'.

Multiplexing:

- In applications using PCM, It is natural to multiplex different message sources by time-division whereby each source keeps its individually throughout the journey from the transmitter to receiver.
- This individuality accounts for the comparative ease with which message source may be dropped or reinserted in a time division multiplex system.